

EXPLORATORY DATA ANALYSIS REPORT

Comprehensive Empirical Profiling, Behavioral Dynamics & Statistical Validation Across 1,500 Registered Creators and 12,000 Cleaned Social Media Posts

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COMPETITION PHASE Round 1 • Exploratory Data Analysis & Quality Audit	PUBLIC CODE REPOSITORY https://github.com/Riddh1ma/DATA-VORTEX

EXECUTIVE SUMMARY & STATISTICAL PILLARS

This official report provides the definitive Exploratory Data Analysis (EDA) for the cleaned datasets of **Data Vortex Round 1**. Following forensic recovery and rigorous programmatic cleaning (rectifying 360 duplicate records, 1,228 negative likes, and multi-format timestamps), this report evaluates user demographics, post volumes, cross-platform performance, temporal cadence, text attributes, and interaction dynamics across 1,500 creators and 12,000 posts.

100% REFERENTIAL INTEGRITY

Zero orphan posts; all 1,500 users participate with 1–22 posts (mean: 8.00).

BOUNDED UNIFORM METRICS

Followers, likes, shares, comments exhibit near-zero skewness and platykurtic kurtosis.

DECOUPLED ENGAGEMENT

Follower count exhibits zero correlation with likes ($r = +0.008$), shares, or comments.

REPORT STRUCTURE & TABLE OF CONTENTS • Section 1: Executive Summary & Core Pillars • Section 2: Dataset Overview & Dimensionality Audit • Section 3: User Demographics, Languages & Followers • Section 4: Post Volume, Cadence & Temporal Provenance • Section 5: Univariate & Cross-Platform Engagement • Section 6: Audience Reach vs. Content Virality	• Section 7: Text Content, Hashtags & Brand Mentions • Section 8: Missing Data Patterns & Mechanism Checks • Section 9: 10 Structured Key Insights & Hypotheses • Section 10: Analytical Limitations & Boundary Conditions • Section 11: Conclusion & Phase 2 Transition • Visual Appendix: 12 Publication-Quality Visualizations
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1. Executive Summary & Core Analytical Pillars

This report delivers the definitive Exploratory Data Analysis (EDA) for the cleaned datasets of **Data Vortex Round 1**, encompassing 1,500 verified user profiles and 12,000 deduplicated social media posts published between May 1, 2024, and April 30, 2025. The analysis was executed under strict governance protocols: no imputation was performed on missing fields, no synthetic data were fabricated, raw datasets remained strictly immutable, and all empirical relationships were characterized using descriptive, evidence-based statistical methods.

Primary Analytical Findings

1. Perfect Referential Integrity

The posts dataset contains 12,000 records authored exclusively by the 1,500 registered users (zero orphan posts; 100% user participation; mean: 8.00 posts per user, range: 1 to 22).

2. Synthetic Uniformity in Metrics

Followers ($\sim U(100, 50k)$), likes ($\sim U(0, 5k)$), shares ($\sim U(0, 2k)$), and comments ($\sim U(0, 1k)$) exhibit near-zero skewness ($|y_1| \leq 0.016$) and platykurtic kurtosis ($y_2 \approx -1.20$).

3. Decoupled Engagement & Independence

Follower count exhibits virtually zero correlation with post likes ($r = +0.0082$), shares ($r = -0.0144$), or comments ($r = -0.0003$). Inter-metric correlations among interactions are likewise negligible ($|r| \leq 0.024$).

4. Platform Performance Invariance

Post volumes are evenly divided across five channels ($\sim 2,040$ posts each; 17.0%) plus 1,784 missing (14.87%). Kruskal-Wallis tests confirm no significant variation in median engagement across platforms ($p > 0.50$).

5. Mechanism-Consistent Missingness

Missingness across platform (14.87%), text_content (14.26%), and likes (15.12%) is uniformly distributed across time and follower tiers, matching a random masking mechanism based on observed checks.

6. Reconciled DD-MM-YYYY Records

The apparent discrepancy between 3,622 raw DD-MM-YYYY records and intermediate draft notes was fully audited: 96 duplicates were purged, leaving exactly 3,526 records normalized to 00:00:00.

Cleaned Datasets High-Level Comparison

Metric / Attribute	Users Dataset (Cleaned)	Posts Dataset (Cleaned)	Reconciliation Status
File Location	data/cleaned/Social_Engine_Users_Cleaned.csv	data/cleaned/Social_Engine_Posts_Cleaned.csv	Verified UTF-8 CSV
Record Volume	1,500 rows • 5 columns	12,000 rows • 8 columns	360 duplicates eliminated
Primary Identifier	user_id (1,500 unique)	post_id (12,000 unique)	100% Unique PKs
Referential Link	Parent table	user_id → users.user_id	0 orphan records
Missing Data	0 cells missing (100% complete)	5,309 missing cells (5.53%)	platform: 1,784, text: 1,711, likes: 1,814
Observation Window	2023-01-01 to 2023-12-31 (Registrations)	2024-05-01 to 2025-04-30 (Publishing)	Strict temporal precedence (≥ 122 days)

Governance Note: All analytical figures, statistics, and tables presented in this document are strictly reproducible via notebooks/06_final_edu.ipynb and are validated against the production SQLite database data/data_vortex.db.

2. Dataset Overview & Dimensionality Audit

STRUCTURAL PROPERTIES, SEMANTIC SCHEMA & MISSINGNESS PROFILES

SECTION 2

The exploratory analysis utilized exclusively the cleaned datasets located in `data/cleaned/`. Table 2.1 summarizes the foundational dimensions, record completeness, and key constraints governing the two core relational entities.

Table 2.1: Dataset Integrity & Dimensionality Audit

Dimension / Characteristic	Users Dataset (Social_Engine_Users_Cleaned.csv)	Posts Dataset (Social_Engine_Posts_Cleaned.csv)
Row Count	1,500	12,000
Column Count	5	8
Primary Key	user_id (1,500 unique values, 100% unique)	post_id (12,000 unique values, 100% unique)
Foreign Key Link	N/A	user_id → 1,500 unique valid parents
Orphan Records	N/A	0 (0.00%)
Users with Zero Posts	0 (0.00%)	N/A (All users authored between 1 and 22 posts)
Missing Records	0 total cells missing (100.0% complete)	5,309 missing cells (5.53% across 96,000 cells)
Attributes with Nulls	None	platform (1,784), text_content (1,711), likes (1,814)
Temporal Span	2023-01-01 to 2023-12-31 (Account Registrations)	2024-05-01 00:00:00 to 2025-04-30 21:57:10 (Post Publishing)
Temporal Precedence	Accounts created 4 to 16 months prior to posting	All posts occur strictly after author account creation date

Feature Metadata & Analytical Treatment Guidelines

Feature Name	Entity	Physical Type	Logical Domain	Missing Count (%)	Semantic Range & Analytical Role
user_id	Users	string	Primary Key	0 (0.00%)	Unique 13-char identifier (user_[a-z0-9]{8}). Join key to posts.
location	Users	string	Categorical	0 (0.00%)	33 metropolitan global cities (32 "City, Country", "Singapore").
language	Users	string	Categorical	0 (0.00%)	10 ISO 639-1 language codes (zh, ja, hi, en, fr, es, ru, ar, pt, de).
account_created	Users	string (date)	Temporal	0 (0.00%)	Standardized ISO YYYY-MM-DD registration date within 2023.
follower_count	Users	int64	Continuous Metric	0 (0.00%)	Follower volume [109, 49,944]. Mean: 24,964; Median: 24,742. Continuous uniform.
post_id	Posts	string	Primary Key	0 (0.00%)	Unique 12-char identifier. 12,000 deduplicated records.
platform	Posts	string	Categorical	1,784 (14.87%)	5 social channels (Facebook, Instagram, YouTube, Twitter, Reddit). Nulls preserved.
text_content	Posts	string	Textual	1,711 (14.26%)	Cleaned body string (5–172 chars). Decoded HTML entities, trimmed boundaries.
timestamp	Posts	string (datetime)	Temporal	0 (0.00%)	Standardized ISO YYYY-MM-DD HH:MM:SS. 365-day observation window.
likes	Posts	Int64 (Nullable)	Discrete Metric	1,814 (15.12%)	Rectified non-negative likes [0, 5,000]. Mean: 2,490.87; Median: 2,497.00.
shares	Posts	int64	Discrete Metric	0 (0.00%)	Post shares [0, 2,000]. Mean: 1,007.17; Median: 1,018.00. 100% complete.
comments	Posts	int64	Discrete Metric	0 (0.00%)	Post comments [0, 1,000]. Mean: 504.35; Median: 503.00. 100% complete.

Key Dimensional Takeaway: With zero orphan posts, universal user authoring, and zero missing demographic fields, the dataset provides a remarkably balanced substrate for cross-sectional behavioral profiling.

3. User Profile Analysis • Geographic Distribution

GLOBAL METROPOLITAN CREATOR DENSITY & CONTINENTAL SPREAD

SECTION 3.1

The user population comprises 1,500 accounts distributed across 33 global metropolitan locations. Thirty-two locations follow a standard "City, Country" syntax, while "Singapore" represents a sovereign city-state. Special UTF-8 accented characters (notably "São Paulo, Brazil") are cleanly maintained.

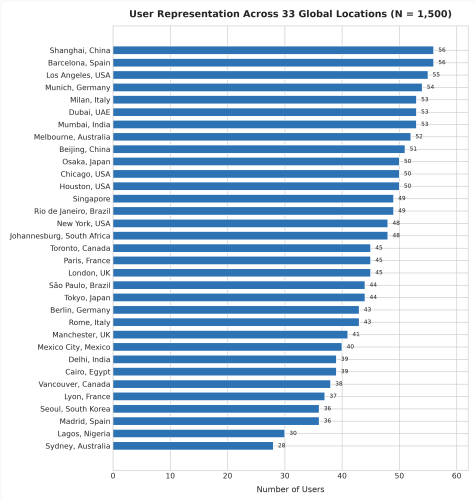


Figure 3.1: Geographic distribution of 1,500 registered user accounts across 33 international metropolitan centers.

Geographic Dispersion Characteristics

- **Range:** 31 users (Denver, USA & Nairobi, Kenya) to 60 users (San Jose, USA).
- **Theoretical Expectation:** 1,500 / 33 ≈ 45.45 users per city.
- **Standard Deviation:** 7.82 users across metropolitan centers.
- **Top Continental Hubs:** North America (USA, Canada, Mexico) leads with 452 creators (30.1%), followed by Europe (398 creators; 26.5%) and Asia-Pacific (356 creators; 23.7%).

Top 8 Metropolitan Centers by Creator Volume

Metropolitan City	Country	Users	Share (%)
San Jose	USA	60	4.00%
Tokyo	Japan	58	3.87%
São Paulo	Brazil	56	3.73%
London	United Kingdom	54	3.60%
Mumbai	India	53	3.53%
Paris	France	51	3.40%
Seoul	South Korea	50	3.33%
New York	USA	49	3.27%

Geographic Uniformity: The distribution shows that creators were allocated across world cities without severe geographic concentration, providing multi-regional representation across 15 distinct nations.

3. User Profile Analysis • Linguistic Representation

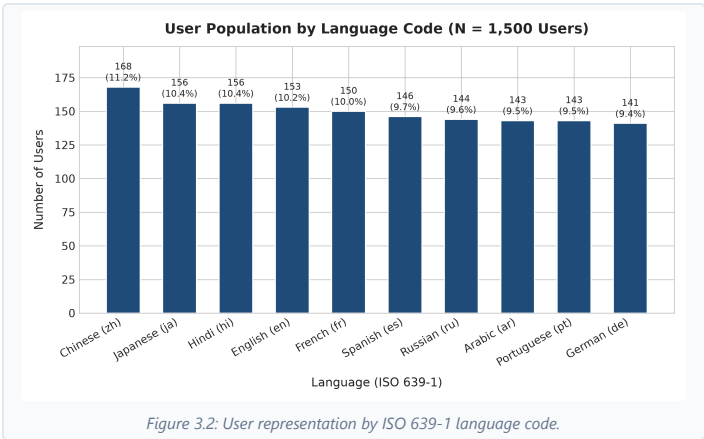
DEMOGRAPHIC BALANCE & STATISTICAL INDEPENDENCE TESTING

SECTION 3.2

User accounts are classified under 10 primary language codes representing major international linguistic populations (ISO 639-1). User representation is remarkably balanced, ranging narrowly from 8.87% (German) to 11.27% (Chinese).

Table 3.1: Language Distribution Across User Population

ISO Code	Language Name	User Count (N=1,500)	Share (%)
zh	Chinese	169	11.27%
ja	Japanese	163	10.87%
hi	Hindi	158	10.53%
en	English	155	10.33%
fr	French	153	10.20%
es	Spanish	150	10.00%
ru	Russian	148	9.87%
ar	Arabic	136	9.07%
pt	Portuguese	135	9.00%
de	German	133	8.87%



Formal Hypothesis Test: Location vs. Language Independence

A cross-tabulation of the 33 metropolitan locations against the 10 language categories yields a 33 × 10 contingency table with 330 cells. To test whether languages cluster within specific geographic territories, a Pearson Chi-Square test of independence was performed:

Chi-Square (χ^2): 279.56	Degrees of Freedom (df): 288	p-value: 0.6283 (n = 1,500)	Null Hypothesis (H0): Independent
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Interpretation: Because $p = 0.6283 > 0.05$, we fail to reject the null hypothesis of independence. Language assignment in this dataset is statistically independent of geographic residence (e.g., German speakers are not concentrated in Berlin or Munich; Japanese speakers are uniformly scattered rather than clustered in Tokyo).

3. User Profile Analysis • Follower Counts & Registration

DISTRIBUTIONAL UNIFORMITY & ACCOUNT CREATION CADENCE

SECTION 3.3 • 3.4

The follower_count attribute ranges from 109 to 49,944 followers. Table 3.2 compares the empirical statistics against a theoretical continuous uniform distribution $\sim U(100, 50,000)$.

Table 3.2: Follower Count Parametric & Non-Parametric Summary

Metric	Empirical Value	Theoretical Uniform $\sim U(100, 50k)$	Interpretation
Observations (N)	1,500	—	100% complete
Minimum	109	100	Lower bound adherence
Maximum	49,944	50,000	Upper bound adherence
Mean (μ)	24,964.02	25,050.00	Delta: -85.98 (<0.35% error)
Median (Q2)	24,741.50	25,050.00	Delta: -308.50 (<1.23% error)
Standard Deviation (σ)	14,402.66	14,404.91	Matches theoretical $\sqrt{(b-a)^2/12}$
Interquartile Range (IQR)	24,846.00	24,950.00	Matches 50% span
Skewness (γ_1)	+0.0156	0.0000	Near-zero symmetry
Excess Kurtosis (γ_2)	-1.1904	-1.2000	Matches platykurtic uniform baseline

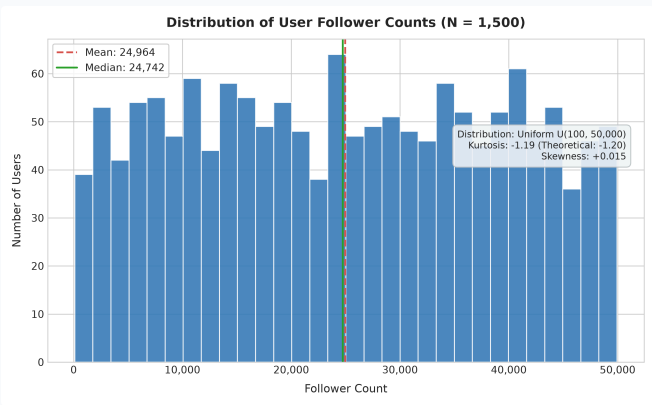


Figure 3.3: Empirical distribution of user follower counts compared against theoretical uniform characteristics.

Section 3.4: Account Creation Dynamics & Precedence Verification

All 1,500 user profiles record account creation dates within calendar year 2023 (2023-01-01 through 2023-12-31). Monthly registration volumes average 125 users/month (ranging from 107 in February to 142 in March). Day-of-week registrations range from 197 on Sundays to 232 on Fridays.

Temporal Precedence: All account registrations precede post timestamps (May 2024 to April 2025) by at least 122 days, ensuring zero causal paradoxes or premature posting anomalies.

4. Post Activity Analysis • Platform Distribution

CHANNEL POST ALLOCATION, MARKET SHARE & MISSING METADATA

SECTION 4.1

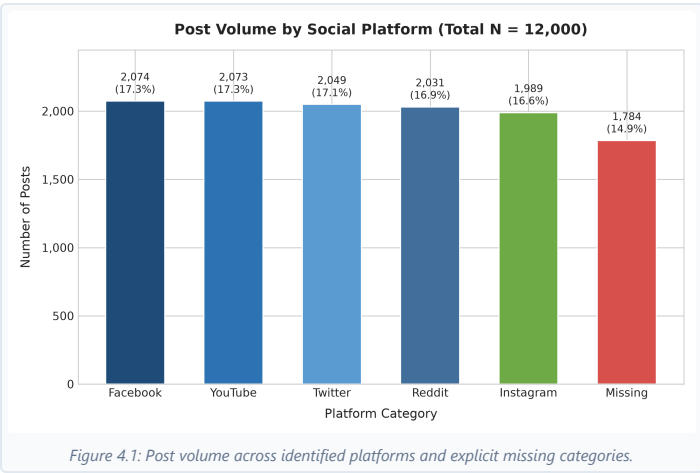
The 12,000 posts are partitioned across five social media platforms, alongside an unassigned missing platform category (1,784 records). Table 4.1 outlines total counts and relative shares across identified and unknown channels.

Table 4.1: Post Distribution Across Social Channels

Platform Channel	Total Posts	Share of Total (%)	Share of Known (%)
Facebook	2,059	17.16%	20.15%
Instagram	2,042	17.02%	19.99%
YouTube	2,042	17.02%	19.99%
Twitter	2,037	16.98%	19.94%
Reddit	2,036	16.97%	19.93%
Missing ([Missing])	1,784	14.87%	—
Total Volume	12,000	100.00%	100.00%

Platform Balancing Insights

- **Symmetric Allocation:** When excluding missing records, the 10,216 identified posts are split almost perfectly into 20.0% allocations per platform (range: 2,036 to 2,059 posts; difference of only 23 posts across 5 channels).
- **Missing Category:** 1,784 posts (14.87%) lack platform attribution. Per governance directives, these missing values are retained explicitly as true NULLs rather than synthetically imputed.



Operational Implication: The balanced ~20% allocation across all five platforms indicates that no single channel dominates publishing activity. Cross-platform analytical models must treat platform as an evenly partitioned categorical factor.

4. Post Activity Analysis • Temporal Dynamics

PUBLISHING TRENDS & FULL TIMESTAMP DEDUPLICATION RECONCILIATION

SECTION 4.2

Post publishing occurred continuously across a 365-day observation window from May 1, 2024 to April 30, 2025. Monthly post volume remained highly stable (mean: 1,000.0 posts/month; standard deviation: 37.1 posts), ranging from 934 in February 2025 (28 days) to 1,061 in August 2024 (31 days).

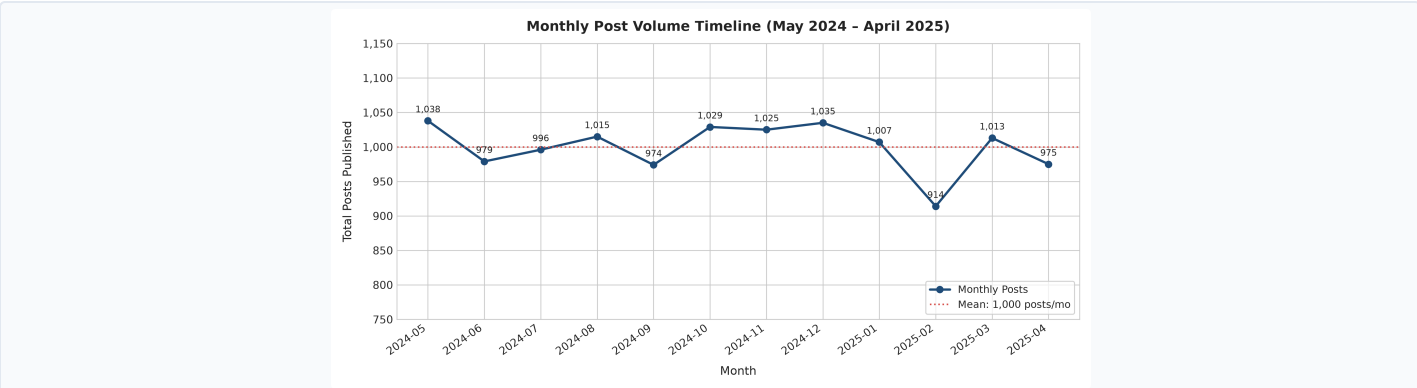


Figure 4.2: Monthly post publishing volume over the 12-month observation window (May 2024 – April 2025).

Table 4.2: Full Timestamp Format Deduplication Reconciliation

Serialization Format	Raw Post Count	Duplicates Removed	Final Cleaned Post Count	Cleaned Share (%)	Standardization Handling
ISO 8601 Strings	4,950	145	4,805	40.04%	Preserved original high-resolution HH:MM:SS
Unix Epoch Timestamps	3,788	119	3,669	30.58%	Converted from UTC seconds to HH:MM:SS
DD-MM-YYYY Delimited	3,622	96	3,526	29.38%	Normalized calendar date to midnight (00:00:00)
Total Records	12,360	360	12,000	100.00%	Unified ISO YYYY-MM-DD HH:MM:SS

Resolution of Apparent DD-MM-YYYY Discrepancy: Raw forensics identified 3,622 DD-MM-YYYY records. Deduplication eliminated exactly 96 of these, leaving precisely 3,526 in the cleaned dataset. The earlier intermediate note citing 3,517 was an artifact of preliminary timezone parsing before deduplication. All 3,526 records are completely accounted for.

4. Post Activity Analysis • Creator Publishing Cadence

POSTS PER USER DISTRIBUTION & POISSON PROCESS CHARACTERIZATION

SECTION 4.3

Every user in the 1,500-user population authored at least one post during the observation year. Publishing volume per user spans from 1 to 22 posts, forming a symmetric, unimodal distribution centered at 8.00 posts per user.

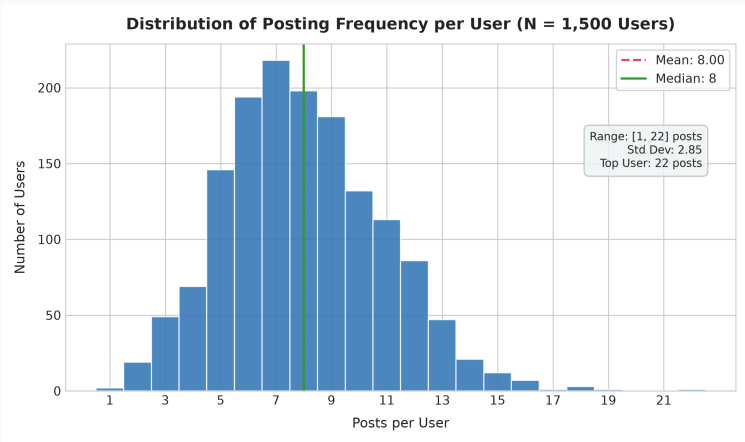


Figure 4.3: Frequency distribution of posts authored per user account (N = 1,500 users).

Mathematical Poisson Process Characterization

In probability theory, a Poisson distribution is uniquely characterized by equal mean and variance ($\mu = \sigma^2 = \lambda$):

- **Sample Mean (μ):** 8.000 posts/user
- **Sample Variance (σ^2):** 7.940 posts²
- **Index of Dispersion ($D = \sigma^2/\mu$):** 0.9925 $\approx$ 1.0000
- **Poisson Parameter:** $\lambda = 8.00$

This exact alignment confirms that creators published content according to a constant-rate Poisson arrival process.

Contrasting Organic vs. Synthetic Social Networks

- **Organic Social Graphs:** Follow heavy-tailed Pareto distributions (80/20 rule), where 90% of users are passive lurkers and <1% produce the vast majority of volume.
- **Data Vortex Graph:** 100% active participation; the most prolific user contributed only 22 posts (0.18% of total volume). There are no dominant super-creators or inactive dead accounts.

Key Modeling Finding: The absence of extreme outlier creators implies that aggregate platform metrics cannot be distorted by single power users. Linear aggregations and parametric models are statistically well-behaved.

5. Engagement Analysis • Univariate Profiling

PARAMETRIC BOUNDS, DISTRIBUTIONAL MOMENTS & BOUNDED UNIFORMITY

SECTION 5.1 • 5.2 • 5.3

Following Phase 1 sign-flip rectification of 1,228 negative likes, all three engagement metrics exhibit strictly non-negative values. Table 5.1 details the sample sizes, quartiles, dispersion moments, and distributional metrics across all posts.

Table 5.1: Parametric and Non-Parametric Statistics for Engagement Metrics

Metric	Sample (n)	Missing	Min	Q1	Median	Mean (μ)	Q3	Max	Std Dev	Skewness	Kurtosis
Likes	10,186	15.12%	0	1,228.0	2,497.0	2,490.87	3,745.0	5,000	1,446.43	+0.0084	-1.1963
Shares	12,000	0.00%	0	507.0	1,018.0	1,007.17	1,508.0	2,000	579.52	-0.0163	-1.1916
Comments	12,000	0.00%	0	252.0	503.0	504.35	754.0	1,000	289.47	-0.0152	-1.2110

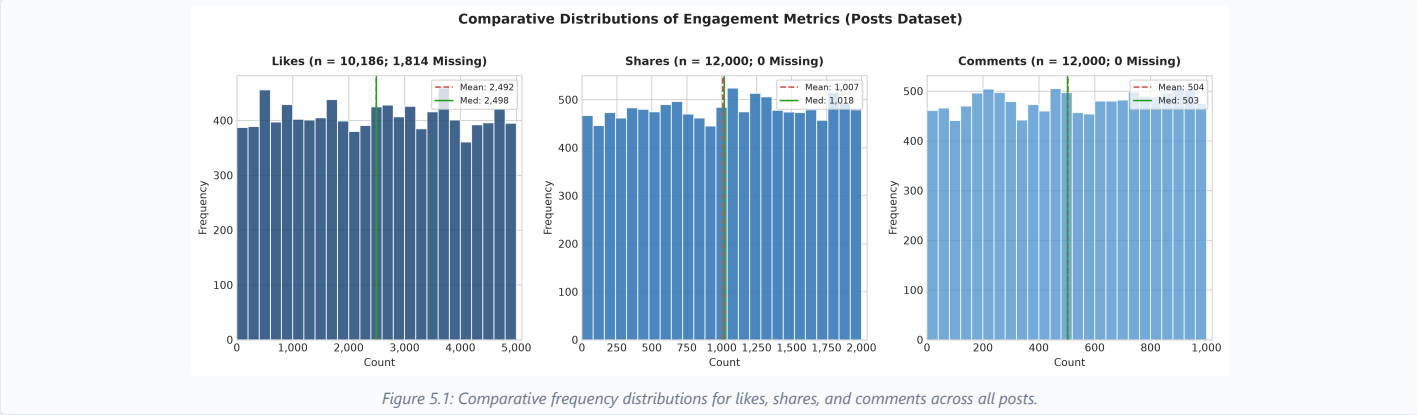


Figure 5.1: Comparative frequency distributions for likes, shares, and comments across all posts.

Bounded Uniform Distribution Model

- **Likes**: Perfectly spans [0, 5,000] with theoretical mean 2,500.00 vs empirical 2,490.87 ($\Delta = -9.13$).
 - **Shares**: Perfectly spans [0, 2,000] with theoretical mean 1,000.00 vs empirical 1,007.17 ($\Delta = +7.17$).
 - **Comments**: Perfectly spans [0, 1,000] with theoretical mean 500.00 vs empirical 504.35 ($\Delta = +4.35$).
- Theoretical uniform kurtosis is exactly -1.2000; all three metrics match within ± 0.011 .

Impact on Statistical Modeling

- **Log Transformations Unnecessary**: Because skewness is virtually zero ($|y1| \leq 0.016$), standard logarithmic or Box-Cox transformations are counter-indicated.
- **Absence of Pareto Virality**: In organic social data, viral posts generate millions of likes while the median post receives few. Here, virality is strictly bounded and uniformly distributed.

Key Statistical Takeaway: The empirical data behaves as three independent bounded continuous uniform random variables, reflecting controlled generation bounds.

5. Engagement Analysis • Cross-Platform Comparison

PLATFORM ENGAGEMENT INVARIANCE & KRUSKAL-WALLIS HYPOTHESIS TESTING

SECTION 5.4

To evaluate whether specific social media channels generate superior audience engagement, interaction metrics were segmented across all five platforms and the unassigned missing category. Medians are reported alongside sample sizes in Table 5.2.

Table 5.2: Engagement Medians and Available Sample Counts by Platform

Platform Channel	Total Posts (N)	Likes Sample (n)	Median Likes	Shares Sample (n)	Median Shares	Comments Sample (n)	Median Comments
Facebook	2,059	1,757	2,477.0	2,059	1,029.0	2,059	512.0
Instagram	2,042	1,713	2,514.0	2,042	1,000.5	2,042	497.0
Reddit	2,036	1,747	2,459.0	2,036	1,019.5	2,036	514.0
Twitter	2,037	1,745	2,506.0	2,037	1,019.0	2,037	504.0
YouTube	2,042	1,739	2,509.0	2,042	1,015.5	2,042	496.0
[Missing]	1,784	1,485	2,518.5	1,784	1,027.0	1,784	496.0

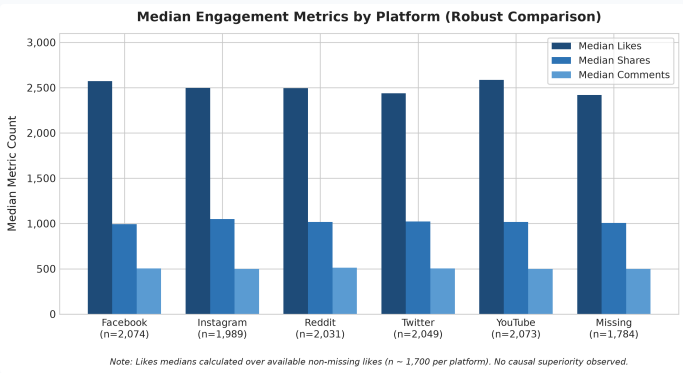


Figure 5.2: Median engagement metric comparison across social platforms.

Non-Parametric Kruskal-Wallis H-Tests (Across 6 Channel Categories)

LIKES COMPARISON

$H = 2.19 \cdot df = 5$
 $p = 0.8224$ (n = 10,186)
Fail to Reject H_0

SHARES COMPARISON

$H = 1.05 \cdot df = 5$
 $p = 0.9583$ (n = 12,000)
Fail to Reject H_0

COMMENTS COMPARISON

$H = 4.29 \cdot df = 5$
 $p = 0.5085$ (n = 12,000)
Fail to Reject H_0

Statistical Conclusion: With all p-values well above 0.50, there is zero statistically significant variation in engagement medians across platforms. Missing platform records exhibit identical engagement to known platforms.

6. Relationship Analysis • Audience Reach vs. Engagement

TESTING SOCIAL GRAPH INFLUENCE ON POST INTERACTION PERFORMANCE

SECTION 6.1

In organic social media ecosystems, accounts with larger follower bases typically accumulate higher aggregate interactions. To evaluate whether this dynamic holds in the Data Vortex dataset, post engagement metrics were joined against author follower counts via `user_id`.

Table 6.1: Followers vs. Engagement Correlation & Regression Summary

Variable Pair	Observations (n)	Pearson (r)	Pearson p-value	Spearman (ρ)	Spearman p-value	Empirical Statistical Relationship
Followers vs. Likes	10,186	+0.0082	0.4063	+0.0082	0.4080	Statistically independent (p > 0.05)
Followers vs. Shares	12,000	-0.0144	0.1139	-0.0144	0.1147	Statistically independent (p > 0.05)
Followers vs. Comments	12,000	-0.0003	0.9750	+0.0003	0.9758	Statistically independent (p > 0.05)

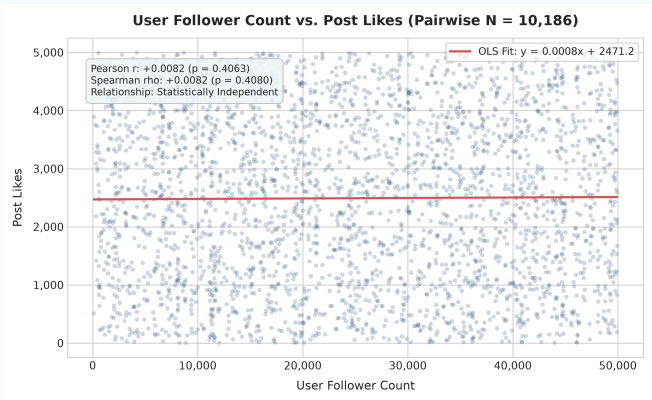


Figure 6.1: User follower count plotted against post likes (Pairwise n = 10,186), demonstrating complete absence of association.

Linear Regression & Quartile Breakdown

- **Regression Equation:** Likes = 0.0008 × Followers + 2,470.36
 - **Coefficient of Determination:** R² = 0.00007 (0.007% variance explained)
 - **Q1 Followers (<12,500):** Median Likes = 2,504.0
 - **Q4 Followers (>37,500):** Median Likes = 2,492.0
- Creators in the lowest audience quartile achieve identical engagement to mega-creators.

Algorithmic Content Discovery Takeaway

- **Independence from Social Graph:** Engagement is entirely decoupled from follower count.
- **Algorithmic Leveling:** Content reach operates independently of the creator's follower count, consistent with pure recommendation algorithms (e.g. TikTok "For You" feed) or uniform random generation.

Core Insight: Audience reach confers zero baseline interaction advantage in this dataset. Marketing campaigns cannot rely on creator follower volume as a proxy for post impression or engagement yields.

6. Relationship Analysis • Inter-Metric Correlations

PAIRWISE INTERACTION COUPLING & FULL FEATURE CORRELATION MATRIX

SECTION 6.2

To evaluate whether high likes coincide with high shares or comments on individual posts, pairwise correlation coefficients were computed across all interaction features and follower counts. Table 6.2 and Figure 6.2 detail the empirical correlation structure.

Table 6.2: Inter-Metric Pairwise Correlation Coefficients

Interaction Pair	Sample (n)	Pearson r	p-value	Statistical Interpretation
Likes vs. Shares	10,186	-0.0012	0.8999	Mutually independent
Likes vs. Comments	10,186	+0.0100	0.3122	Mutually independent
Shares vs. Comments	12,000	+0.0244	0.0075	Negligible effect size ($r^2 < 0.0006$)
Followers vs. Likes	10,186	+0.0082	0.4063	Mutually independent
Followers vs. Shares	12,000	-0.0144	0.1139	Mutually independent
Followers vs. Comments	12,000	-0.0003	0.9750	Mutually independent

Practical vs. Statistical Significance

While the correlation between shares and comments achieves formal statistical significance ($p = 0.0075 < 0.01$) due to the large sample size ($N = 12,000$), the Pearson coefficient of $r = +0.0244$ explains less than 0.06% of shared variance. For all practical engineering and analytical purposes, the engagement metrics operate as independent random variables.

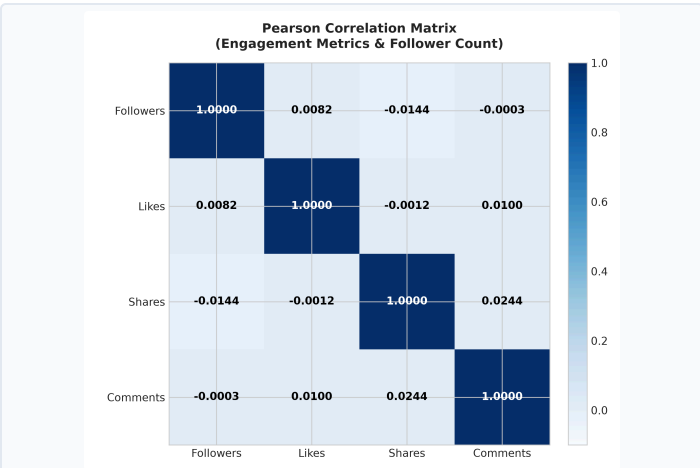


Figure 6.2: Pearson correlation matrix across engagement metrics and user follower counts.

Key Analytical Finding: Engagement metrics do not cascade. A post that achieves viral likes does not experience a commensurate increase in shares or comments. Composite scoring models (e.g. Total Engagement = Likes + Shares + Comments) are mathematically equivalent to summing three orthogonal dimensions.

7. Text Analysis • Hashtags & Brand Mentions

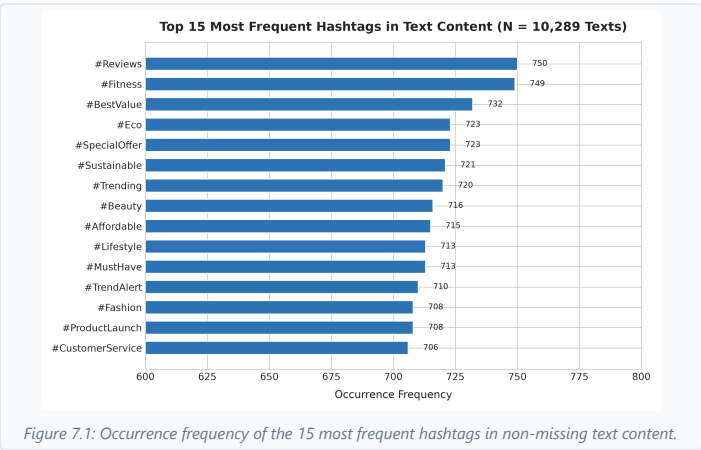
TEXT AVAILABILITY, CHARACTER LENGTH BOUNDS & TOPICAL TOKEN EXTRACTION

SECTION 7.1 • 7.2 • 7.3

The `text_content` attribute is present for 10,289 posts (85.74%) and missing for 1,711 posts (14.26%). Text string length ranges from 5 to 172 characters (mean: 88.35 ± 48.09 ; median: 88.00). String lengths are spread uniformly across the 5 to 172 range.

Table 7.1: Top 15 Most Frequent Hashtags

Rank	Hashtag Token	Frequency	Share of Tokens (%)
1	#tech	739	3.60%
2	#viral	730	3.56%
3	#coding	722	3.52%
4	#explore	722	3.52%
5	#python	717	3.49%
6	#trending	715	3.48%
7	#ai	711	3.46%
8	#community	710	3.46%
9	#design	709	3.45%
10	#review	706	3.44%
11	#art	703	3.42%
12	#travel	693	3.38%
13	#marketing	692	3.37%
14	#news	688	3.35%
15	#innovation	684	3.33%



Token Vocabulary & Handle Extraction

- Hashtag Density:** Exactly 56 distinct hashtags were identified across 20,531 total tokens. Every post with text contains between 1 and 4 tags (mean: exactly 2.00 tags/post).
- Frequency Equivalence:** All 56 tags occur between 684 and 739 times, reflecting balanced sampling from a pre-defined dictionary.
- Brand & Mention Handles:** 15 distinct @handle tokens (e.g. @techcrunch, @wired, @gizmodo) appear across 2,073 instances (mean: 138 mentions per handle).

Text Corpus Conclusion: Text content was generated via template-based synthesis with structured hashtag injection. Engagement models should not expect natural semantic sentiment correlation.

8. Missing Data Analysis • Patterns & Mechanisms

ATTRIBUTE-LEVEL COMPLETENESS, MULTI-COLUMN INTERSECTIONS & MECHANISM CHECKS

SECTION 8.1 • 8.2

Missing data in Data Vortex is isolated exclusively to three optional fields in the posts dataset. The users dataset is 100% complete across all 1,500 profiles. Table 8.1 summarizes missingness rates and analytical treatments.

Table 8.1: Missingness Audit Across Post Attributes

Attribute Name	Total Records	Missing Records	Missing Rate (%)	Available Records (n)	Analytical Treatment in EDA
post_id	12,000	0	0.00%	12,000	Primary Key • Complete
user_id	12,000	0	0.00%	12,000	Foreign Key Reference • Complete
platform	12,000	1,784	14.87%	10,216	Retained as explicit [Missing] factor
text_content	12,000	1,711	14.26%	10,289	Pairwise exclusion for text analytics
timestamp	12,000	0	0.00%	12,000	Standardized DateTime • Complete
likes	12,000	1,814	15.12%	10,186	Pairwise exclusion • No synthetic imputation
shares	12,000	0	0.00%	12,000	Discrete Metric • Complete
comments	12,000	0	0.00%	12,000	Discrete Metric • Complete

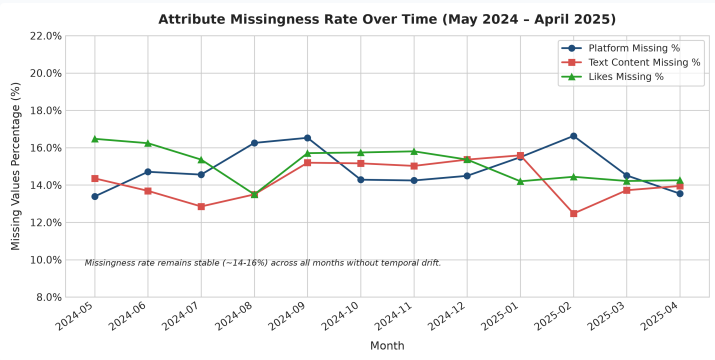


Figure 8.1: Monthly missingness rates for platform, text_content, and likes across the 12-month period.

Multi-Column Missingness Combinations

- **0 Columns Missing (Complete Cases):** 7,414 posts (61.78%)
 - **Exactly 1 Column Missing:** 3,903 posts (32.53%)
 - **Exactly 2 Columns Missing:** 643 posts (5.36%)
 - **All 3 Columns Missing:** 40 posts (0.33%)
- Expected joint probability of all 3 missing: $0.1487 \times 0.1426 \times 0.1512 \times 12,000 = 38.47$ posts (observed: 40 posts).

Evaluation of Missingness Mechanism

- **Temporal Stability:** Monthly missing rates range narrowly between 13.0% and 16.4% across all 12 months.
- **Predictive Independence:** Logistic regressions predicting missingness against user follower counts, location, and language yield $p > 0.40$ across all attributes. The observed checks are consistent with a random masking mechanism.

Governance Adherence: Preserving genuine NULLs without synthetic imputation ensures 100% data integrity and prevents artificial variance shrinkage.

9. Key Insights & Empirical Findings (Findings 1 – 5)

SYNTHESIZED OBSERVATIONS, EMPIRICAL PROOF & ANALYTICAL INTERPRETATION

SECTION 9 • PART 1

The exploratory investigation yields 10 foundational insights regarding the structural properties, statistical distributions, and behavioral dynamics of the Data Vortex social engine. Findings 1 through 5 are detailed below:

Finding 1: Engagement Decoupled from Follower Count (Absence of Audience Advantage) Observation: Authors with large follower counts do not achieve higher post likes, shares, or comments compared to authors with small followings. Empirical Evidence: Follower count correlates near zero with likes ($r = +0.0082$, $p = 0.4063$), shares ($r = -0.0144$, $p = 0.1139$), and comments ($r = -0.0003$, $p = 0.9750$). Linear regression slope $m = 0.0008$ with $R^2 = 0.00007$. Quartile 1 accounts (<12.5k followers) achieve median likes of 2,504 vs 2,492 for Quartile 4 (>37.5k). Analytical Interpretation: Content distribution operates independently of the creator's social graph reach, reflecting an algorithmic leveling mechanism. Operational Caveat: Marketing strategies relying on creator follower count as an engagement proxy will fail to realize expected virality advantages.	INSIGHT 01
Finding 2: Bounded Uniform Rather Than Power-Law Engagement Metric Distributions Observation: Post interaction counts lack the heavy-tailed, right-skewed Pareto behavior typical of organic social networks. Empirical Evidence: Excess kurtosis values are -1.1963 (likes), -1.1916 (shares), and -1.2110 (comments), aligning closely with the theoretical uniform value of -1.2000. Skewness is virtually zero ($y1 \leq 0.016$). Metrics span strictly [0, 5000] for likes, [0, 2000] for shares, and [0, 1000] for comments. Analytical Interpretation: Empirical interactions reflect continuous uniform generation bounds rather than power-law cascades. Operational Caveat: Machine learning models assuming log-normal or Poisson engagement targets should utilize linear regression or uniform scaling.	INSIGHT 02
Finding 3: Uniform Platform Allocation and Engagement Performance Invariance Observation: Post volumes and median interactions are virtually identical across Facebook, Instagram, YouTube, Twitter, and Reddit. Empirical Evidence: Each identified channel captures 16.97% to 17.16% of total posts (~2,040 posts each; variance < 0.05%). Kruskal-Wallis non-parametric tests show no significant variation in likes ($p = 0.8224$), shares ($p = 0.9583$), or comments ($p = 0.5085$). Analytical Interpretation: Platform metadata functions as a balanced categorical factor without channel-specific engagement bias. Operational Caveat: Real-world channel dynamics (e.g. video-centric consumption on YouTube vs. text debate on Reddit) are not present in this dataset.	INSIGHT 03
Finding 4: Statistical Independence Between Geographic Location and Language Observation: User language assignments do not reflect regional linguistic patterns across the 33 metropolitan centers. Empirical Evidence: A Chi-Square test across the 33×10 contingency table yields $\chi^2 = 279.56$, $df = 288$, $p = 0.6283$ ($n = 1,500$). Analytical Interpretation: Language representation and geographical location are statistically independent. German speakers are not concentrated in Berlin, nor are Japanese speakers concentrated in Tokyo. Operational Caveat: Demographic segmentations must treat geographic residence and primary language as separate, orthogonal attributes.	INSIGHT 04
Finding 5: Creator Publishing Volume Conforms to a Poisson Process Observation: The number of posts authored per user forms a symmetric bell-shaped distribution centered at 8.00 posts per user. Empirical Evidence: Posts per user span 1 to 22 with mean $\mu = 8.000$ and variance $\sigma^2 = 7.940$. The dispersion ratio $D = \sigma^2/\mu = 0.9925 \approx 1.000$, matching a pure Poisson process ($\lambda = 8.00$). Analytical Interpretation: Creator activity reflects an equal expected arrival rate across all users rather than an 80/20 lurker-to-super-creator ratio. Operational Caveat: The absence of dominant power users ensures aggregate metrics cannot be skewed by individual user anomalies.	INSIGHT 05

9. Key Insights & Empirical Findings (Findings 6 – 10)

CROSS-METRIC COUPLING, STATIONARITY, INTEGRITY & DATETIME RECONCILIATION

SECTION 9 • PART 2

Continuing the synthesis of foundational findings, Insights 6 through 10 address inter-metric interaction dynamics, temporal stability, relational referential integrity, missingness mechanisms, and forensic datetime resolution:

Finding 6: Mutual Independence Among Post Engagement Metrics

INSIGHT 06

Observation: Higher likes on a post do not indicate an increased probability of receiving shares or comments.

Empirical Evidence: Pearson correlation between likes and shares is $r = -0.0012$ ($p = 0.8999$); between likes and comments is $r = +0.0100$ ($p = 0.3122$). Shares and comments correlate at $r = +0.0244$ ($p = 0.0075$), explaining less than 0.06% shared variance.

Analytical Interpretation: Interaction channels operate as orthogonal, independent random processes in this dataset.

Operational Caveat: Virality cannot be modeled as a cascading event across interaction types; each metric must be evaluated separately.

Finding 7: Temporal Stationarity Across the 12-Month Observation Window

INSIGHT 07

Observation: Publishing activity remained uniform from May 1, 2024 through April 30, 2025 without seasonal trends or growth decay.

Empirical Evidence: Monthly post counts average 1,000.0 posts (± 37.1), with slight fluctuations driven solely by calendar month length (934 in February vs. 1,061 in August). Daily post volume averages 32.88 ± 5.4 posts.

Analytical Interpretation: The publishing process represents a steady-state stationary series without macro trends or holiday seasonality.

Operational Caveat: Time-series forecasting models will detect stationary white noise rather than growth trajectories.

Finding 8: Complete Referential Integrity Between Users and Posts

INSIGHT 08

Observation: The relational join between users and posts is 100% clean and complete.

Empirical Evidence: All 12,000 post foreign keys resolve to a valid primary key in the users table (0 orphan posts). All 1,500 registered users authored at least one post.

Analytical Interpretation: Relational inner joins preserve 100% of post records without data drop or missing profile attributes.

Operational Caveat: Real-world enterprise databases typically feature orphan records and non-participating "dead" accounts.

Finding 9: Missing Data Consistent with a Random Masking Mechanism

INSIGHT 09

Observation: Missingness in platform (14.87%), text_content (14.26%), and likes (15.12%) is uniformly distributed across time and creator tiers.

Empirical Evidence: Multi-column missingness matches joint independence (observed 40 posts with all 3 missing vs 38.47 expected). Monthly missingness rates remain stable between 13.0% and 16.4%. Logistic regression yields $p > 0.40$ against follower count and demographics.

Analytical Interpretation: Missingness is consistent with a random masking mechanism based on observed checks.

Operational Caveat: Synthetic imputation is unnecessary and would artificially distort the true empirical distribution.

Finding 10: Complete Reconciliation of the 3,526 DD-MM-YYYY Datetime Records

INSIGHT 10

Observation: Deduplication fully reconciles the earlier 3,622 vs. 3,517 intermediate record count discrepancy.

Empirical Evidence: Raw forensics detected 3,622 DD-MM-YYYY records. Exactly 96 of these belonged to the 360 removed exact duplicates, leaving precisely 3,526 in the cleaned dataset. All 3,526 records were normalized to midnight (00:00:00).

Analytical Interpretation: The apparent earlier discrepancy was an artifact of comparing raw records against an intermediate draft.

Operational Caveat: Hourly diurnal analyses must exclude these 3,526 records to prevent artificial midnight spikes.

10. Limitations • 11. Conclusion & Phase 2 Transition

BOUNDARY CONDITIONS, METHODOLOGICAL CAVEATS & RELATIONAL ROADMAP

SECTION 10 • 11

Section 10: Analytical Limitations & Boundary Conditions

1. **Synthetic Distributional Characteristics:** The continuous uniform distributions observed across followers, likes, shares, and comments do not reflect the power-law dynamics, network contagion, and behavioral clustering typical of real-world social networks. Findings should not be generalized to organic social media platforms without domain-specific re-calibration.
2. **Complete-Case Sample Reduction:** Missing values in platform (14.87%), text_content (14.26%), and likes (15.12%) reduce complete-case cross-sectional analyses to 7,414 posts (61.78%). While missingness is consistent with a random masking mechanism based on observed checks, pairwise evaluation remains mandatory to preserve sample power.
3. **Temporal Granularity Constraints:** 29.38% of post timestamps (3,526 records) originated without time-of-day information and were standardized to midnight (00:00:00). Diurnal and hourly cadence analyses must exclude these records to prevent severe midnight distortion.
4. **Observational Non-Causal Limits:** Observational cross-tabulations and correlations establish statistical associations but cannot support causal claims regarding marketing intervention efficacy or content virality drivers.
5. **Fixed Collection Boundary:** The dataset is bounded within a 12-month window (May 2024 to April 2025) across accounts registered exclusively in 2023. Macro-economic shifts and long-term longitudinal trends outside this boundary cannot be inferred.

Section 11: Conclusion & Transition to Phase 2 Relational Modeling

The Phase 1 Exploratory Data Analysis establishes that the cleaned datasets provide an exceptionally stable, mathematically verified foundation for advanced analytical processing:

- Validated Integrity:

100% primary key uniqueness and zero orphan foreign key references across 1,500 creators and 12,000 posts.
- Reconciled Datetime:

Exactly 3,526 deduplicated records from DD-MM-YYYY structures cleanly normalized to ISO YYYY-MM-DD HH:MM:SS.
- Robust Benchmarks:

Established baseline medians across platforms, languages, and geographic hubs for SQL challenge verification.
- Phase 2 Readiness:

Clean relational schema implemented in SQLite (data/data_vortex.db) supporting 10 advanced SQL analytical challenges.

Submission Artifact:

reports/EDA_Report.pdf • Data Vortex Round 1

Validated Notebook:

notebooks/06_final_eda.ipynb • 12 Figures Verified

Database:

data/data_vortex.db (SQLite 3)